Skill Arena - Phase 7 Document 3

AI Solver Framework, Human Performance Modeling, Difficulty Calibration & Challenge Balancing Architecture

Document Purpose

Define how Skill Arena predicts challenge difficulty, models human performance and balances house challenges over time.

Business Objective

Create a mathematically controlled challenge ecosystem that remains fair, scalable and sustainable.

Core Philosophy

Every challenge should be measurable, predictable and continuously calibrated using real performance data.

Core Components

AI Solver Engine, Human Performance Model, Difficulty Calibration Engine, Challenge Analytics Engine and House Balance Engine.

AI Solver Engine

Every generated maze is solved automatically before being released to players.

Solver Responsibilities

Calculate optimal route, shortest path, decision complexity, trap exposure and theoretical completion time.

Solver Metadata

Optimal Time, Complexity Score, Decision Density, Risk Score and Solver Confidence.

Human Performance Model

Transforms solver results into realistic human completion expectations.

Human Performance Categories

Beginner, Intermediate, Advanced, Expert and Elite player profiles.

Expected Completion Bands

Generate expected completion ranges rather than single completion times.

Difficulty Calibration Engine

Continuously adjusts challenge parameters using live player performance data.

Calibration Inputs

Completion Times, Failure Rates, Retry Rates, Abandonment Rates and Replay Analysis.

Difficulty Score Framework

Every maze receives a dynamic difficulty score that can be compared globally.

Challenge Balancing Engine

Ensures challenge outcomes remain aligned with platform objectives.

House Balance Model

Uses actual player performance and historical outcomes to maintain sustainable challenge economics.

Target House Range

Maintain approximately 60 to 70 percent house success rate with 65 percent as the operating target.

Adaptive Adjustment Logic

Difficulty recommendations generated when player success rates drift outside target ranges.

Performance Outlier Detection

Identify unusually easy, unusually difficult or potentially exploitable challenge structures.

Replay Intelligence Integration

Replay analysis used to refine solver assumptions and human performance models.

Category Calibration

Each category maintains its own performance profile and balancing metrics.

Progression Calibration

Difficulty increases based on measured complexity rather than arbitrary level counts.

Treasury Integration

Challenge balancing considers sustainability and treasury exposure.

Analytics Integration

Performance data contributes to forecasting, balancing and platform intelligence systems.

Monitoring Requirements

Track completion distributions, solver accuracy, house performance and challenge health.

Audit Requirements

All calibration decisions, solver outputs and balancing actions permanently logged.

Future Expansion

Supports AI-generated challenge types, additional games and adaptive difficulty systems.

Acceptance Criteria

Platform can predict challenge outcomes, balance difficulty and maintain sustainable competition at scale.